

[Homework 1] Review of Probability Theory

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Probability Space of Tossing Coins

1. For this problem we need to prove two things: first, each set C_s is a subset of Ω , and second, each $\omega \in \Omega$ is in one and only one C_s .

The first thing is obvious since C_s is defined to be a set composed of ω . Then we also know that ω in each C_s must be different at the first n elements. For any ω , it's first n elements must be in $\{0, 1\}^n$ so it's in one of the C_s s.

According to the definition of partition, $\{C_s\}$ is a partition of Ω .

2. Each C_s has the same property, the σ -algebra of $\{C_s\}$ is $2^{\{C_s\}}$ while we can simple build a bijection from $\{0, 1\}^n$ to $\{C_s\}$. Thus we can build a bijection between $2^{\{C_s\}}$ (or) $\mathcal{F}_n$ and $2^{\{0,1\}^n}$.
3. First prove that $\mathcal{F}_n \in \mathcal{F}_{n+1}$. Any set in $\mathcal{F}_n$ can be obtained with C_{s_n} s. We also know that each C_{s_n} is a union of two $C_{s_{n+1}}$ s. Thus each set in $\mathcal{F}_n$ is in $\mathcal{F}_{n+1}$. However, we can't find a method to obtain a $C_{s_{n+1}}$ with C_{s_n} s. $\mathcal{F}_n \subsetneq \mathcal{F}_{n+1}$
4. For any set $A \in \mathcal{F}_n$ in $\mathcal{F}_\infty$, since $\mathcal{F}_n$ is σ -algebra, A^c is in $\mathcal{F}_n$ and thus A^c is in $\mathcal{F}_\infty$. For $A \in \mathcal{F}_{n_1}$ and $B \in \mathcal{F}_{n_2}$. Let $n = \max\{n_1, n_2\}$, suppose it's n_1 . B can be the union of sets in $\mathcal{F}_{n_1}$. $A \cap B$ is in $\mathcal{F}_{n_1}$, and is in $\mathcal{F}_\infty$.

Then prove $\mathcal{F}_\infty \neq \Omega$. Let A be an all 1 sequence. No matter how big n is, there can't be a $C_{s_n} = A$. Since $\{C_s\}$ is a partition, we can't get A through intersection. A is not in any of $\mathcal{F}_n$. A is in 2^Ω but A is not in $\mathcal{F}_\infty$.

5. $\mathcal{B}(\Omega)$ changes the original version of s with fixed first n position to some m of the first n position. In the proof of the previous problem, the set A can be a one-element set containing any single $\omega \in \Omega$. That is, $\{\omega\}$ is not in $\mathcal{F}_\infty$. Now prove $\{\omega\} \in \mathcal{B}(\Omega)$.

For a specific ω_0 and any ω' other than ω_0 , all the ω' forms a set Ω' , which is also $\{\omega\}^c$. WLOG, we still let ω be an all 1 sequence. For any ω' , find the first 0 at position m , and there is a corresponding C_{s_m} in $\mathcal{F}_m$, ω not in C_{s_m} but ω' in it. The universal union of all such C_{s_m} forms Ω' . $\Omega' \in \mathcal{B}(\Omega)$, and thus proof ends.

6. We can trace back to find that this A is from a specific $\mathcal{F}_a$. Then existence is obvious: when we construct $\mathcal{F}_a$, A is either a C_s (one set, $b = 1$) or a union of C_s s (b sets in total) (remember intersection is empty). Now for this existing a, b , since we can union two $C_{s_{a+1}}$ s to form a C_{s_a} , we can use $2 \times b$ sets from the next set group. We can further use the next α set group, but we have $\frac{k}{2^n} = \frac{2^\alpha b}{2^{a+\alpha}} = \frac{b}{2^a}$ is constant.
7. We first prove $P(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} P(A_n)$. We decompose the set to the same n . This is to say, we can obtain these sets from a single $\mathcal{F}_n$. Then since these sets are exclusive, the additivity holds.

Then, according to Carathéodory's extension theorem, $\mathcal{B}(\Omega) = \sigma(\mathcal{F}_\infty)$, P is a σ -finite measure on $\mathcal{F}_\infty$, sets in $\mathcal{F}_\infty$ is measurable, we claim P is a unique probability measure.

8. $\text{Geom}(1/2)$ can be simply explained as tossing the coin until it's head is up. We claim this event $A_k = \{\omega \in \Omega | w_k = 1; \forall i, w_i = 0\}$. $Pr[k < K] = (1-p)^{k-1}p, p = \frac{1}{2}$. The distribution space is defined as $(\mathbb{R}, \{A_k\}_\infty, Pr)$.

Conditional Expectation

1. Since f is measurable, for any t , set $f^{-1}((-\infty, t])$ is Borel set. $w \in \Omega : f(X(w)) \leq t$ is in $\sigma(X)$: $f(X)^{-1}((-\infty, t]) \in \sigma(X)$. For any Borel set A , $f(X)^{-1}(A) \in \sigma(X)$, and thus $f(X)$ is $\sigma(X)$ -measurable.
2. We have $\sigma(Y) = \sigma(Y')$, $\mathbf{E}[X | Y] = \mathbf{E}[X | \sigma(Y)]$, $\mathbf{E}[X | Y'] = \mathbf{E}[X | \sigma(Y')]$, $\mathbf{E}[X | \sigma(Y)] = \mathbf{E}[X | \sigma(Y')]$. Finally we get $\mathbf{E}[X | Y] = \mathbf{E}[X | Y']$.
3. Recall two properties:
 - $\mathbf{E}[\mathbf{E}[X|\mathcal{F}]] = \mathbf{E}[X]$
 - If $\mathcal{G} \subseteq \mathcal{F}$, then $\mathbf{E}[\mathbf{E}[X|\mathcal{F}]|\mathcal{G}] = \mathbf{E}[X|\mathcal{G}]$

Note we have $\mathcal{F}_1 \subseteq \mathcal{F}_2$. Therefore, $\mathbf{E}[\mathbf{E}[X|\mathcal{F}_1]|\mathcal{F}_2] = \mathbf{E}[X|\mathcal{F}_2]$. Again, but this time with $\mathcal{F}_2 \subseteq \mathcal{F}_1$, we have: $\mathbf{E}[\mathbf{E}[X|\mathcal{F}_2]|\mathcal{F}_1] = \mathbf{E}[X|\mathcal{F}_1]$. This shows that $\mathbf{E}[\mathbf{E}[X|\mathcal{F}_1]|\mathcal{F}_2] = \mathbf{E}[\mathbf{E}[X|\mathcal{F}_2]|\mathcal{F}_1]$. Combining with property 1, we obtain: $\mathbf{E}[\mathbf{E}[X|\mathcal{F}_1]|\mathcal{F}_2] = \mathbf{E}[\mathbf{E}[X|\mathcal{F}_2]|\mathcal{F}_1] = \mathbf{E}[X]$. Therefore, we have $\mathbf{E}[\mathbf{E}[X|\mathcal{F}_1]|\mathcal{F}_2] = \mathbf{E}[\mathbf{E}[X|\mathcal{F}_2]|\mathcal{F}_1] = \mathbf{E}[X|\mathcal{F}_1]$.